AFFINENESS OF THE LIMIT DESCENDS TO A FINITE STAGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1375-1432, following the statement at 1369-1373.

Editorial note. No load-bearing omissions; the closing parenthesis notes X_i = Spec(A_i) then follows but is not needed.

Allowed standard prerequisites (editorial).

- Standard etale presentation of etale maps of affines. Algebra, Lemma algebra-lemma-etale-standard-smooth (cross-chapter, cited at ~1385-1386). Writes B = A[x_1,...,x_n]/(g_1,...,g_n) with invertible Jacobian determinant.
- Descent of sections along the limit. lemma-descend-section (same chapter, cited at ~1387-1388). A = colim A_i.
- Descent of surjectivity. Limits, Lemma limits-lemma-descend-surjective (cross-chapter, cited at ~1397-1399). Spec(B_i) -> Spec(A_i) surjective after increasing i.
- Equality of morphisms descends. lemma-descend-equality (same chapter, cited at ~1421-1423). f_0 becomes an isomorphism after base change to some X_i.
- Descent of affineness along fppf coverings. Descent, Lemmas descent-lemma-descent-data-sheaves and descent-lemma-affine (cross-chapter, cited at ~1426-1428). Concludes X_i -> Spec(A_i) is representable by an affine scheme.

Source reference. Original reference 00U9: focused-seeds/geometry/S9-spaces-limits-limit-is-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0245: focused-seeds/geometry/S9-spaces-limits-limit-is-affine/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02W5: focused-seeds/geometry/S9-spaces-limits-limit-is-affine/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 084S: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1313–1313. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SI: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1259–1259. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SN: focused-seeds/geometry/S9-spaces-limits-limit-is-affine/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084R: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1250–1250. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Situation 084R. If X is affine, then there exists an i such that X_i is affine.*

Proof. Choose $0 \in I$. Choose an affine scheme U_0 and a surjective étale morphism $U_0 \to X_0$. Set $U = U_0 \times_{X_0} X$ and $U_i = U_0 \times_{X_0} X_i$ for $i \geq 0$. Since the transition morphisms are affine, the algebraic spaces U_i and U are affine. Thus $U \to X$ is an étale morphism of affine schemes. Hence we can write $X = \mathrm{Spec}(A)$, $U = \mathrm{Spec}(B)$ and

$$B = A[x_1, \ldots, x_n]/(g_1, \ldots, g_n)$$

such that $\Delta = \det(\partial g_\lambda / \partial x_\mu)$ is invertible in B, see Algebra, Lemma 00U9. Set $A_i = \mathcal{O}_{X_i}(X_i)$. We have $A = \mathrm{colim}\, A_i$ by Lemma 07SI. After increasing 0 we may assume we have $g_{1,i}, \ldots, g_{n,i} \in A_i[x_1, \ldots, x_n]$ mapping to $g_1, \ldots, g_n$. Set

$$B_i = A_i[x_1, \ldots, x_n]/(g_{1,i}, \ldots, g_{n,i})$$

for all $i \geq 0$. Increasing 0 if necessary we may assume that $\Delta_i = \det(\partial g_{\lambda,i} / \partial x_\mu)$ is invertible in B_i for all $i \geq 0$. Thus $A_i \to B_i$ is an étale ring map. After increasing 0 we may assume also that $\mathrm{Spec}(B_i) \to \mathrm{Spec}(A_i)$ is surjective, see Limits, Lemma 07SN. Increasing 0 yet again we may choose elements $h_{1,i}, \ldots, h_{n,i} \in \mathcal{O}_{U_i}(U_i)$ which map to the classes of $x_1, \ldots, x_n$ in $B = \mathcal{O}_U(U)$ and such that $g_{\lambda,i}(h_{\nu,i}) = 0$ in $\mathcal{O}_{U_i}(U_i)$. Thus we obtain a commutative diagram

$$\begin{array}{ccc} X_i & \longleftarrow & U_i \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A_i) & \longleftarrow & \mathrm{Spec}(B_i) \end{array} \tag{0.1.1}$$

By construction $B_i = B_0 \otimes_{A_0} A_i$ and $B = B_0 \otimes_{A_0} A$. Consider the morphism

$$f_0 : U_0 \longrightarrow X_0 \times_{\mathrm{Spec}(A_0)} \mathrm{Spec}(B_0)$$

This is a morphism of quasi-compact and quasi-separated algebraic spaces representable, separated and étale over X_0. The base change of f_0 to X is an isomorphism by our choices. Hence Lemma 084S guarantees that there exists an i such that the base change of f_0 to X_i is an isomorphism, in other words the diagram (0.1.1) is cartesian. Thus Descent, Lemma 02W5 applied to the fppf covering $\{\mathrm{Spec}(B_i) \to \mathrm{Spec}(A_i)\}$ combined with Descent, Lemma 0245 give that $X_i \to \mathrm{Spec}(A_i)$ is representable by a scheme affine over $\mathrm{Spec}(A_i)$ as desired. (Of course it then also follows that $X_i = \mathrm{Spec}(A_i)$ but we don't need this.) □